

DATABASE DESIGN - DRIFTDATER

The schema consists of the following tables:

Table	Key Fields
users	User_id, first_name, last_name, username, dob, looking_for, password, email, gender, has_changed_dob
profiles	profile_id, user_id, visibility, education, photo_url, bio, location, interests
matches	id, user1_id, user2_id, status, matched_at
messages	message_id, match_id, sender_id, receiver_id, content, timestamp
likes	like_id, liker_id, liked_id, created_at, is_match, action
preferences	pref_id, user_id, gender_pref, education_pref, religion_pref, age_min, age_max
savedprofile	id, saver_id, saved_id, timestamp
notifications	notif_id, user_id, message

user

user_id, first_name, last_name, username, dob, looking_for, password, email, gender, has_changed_dob

profile

profile_id, user_id, visibility, education, photo_url, bio, location, interests

preferences

pref_id, user_id, gender_pref, education_pref, religion_pref, age_min, age_max

like

like_id, liker_id, liked_id, created_at, is_match, action

matches

match_id , user1_id, user2_id, matched_at

messages

message_id , match_id, sender_id, receiver_id, content, timestamp

savedprofile

id, saver_id, saved_id, timestamp

notification

notif_id , user_id, message

5.1 Key Relationships

Users → Profiles

One-to-One: each user has exactly one profile.

Users → Notifications

One-to-Many: a user can have multiple notifications.

Users → SavedProfiles

One-to-Many (self-referential join): users can save multiple other users.

Users → Preferences

One-to-One (intended): each user has one preference record.

Users → Likes

Many-to-Many (self-referential, directed): users can like many users and be liked by many users.

Users → Matches

Many-to-Many (self-referential, derived): matches are created from mutual likes.

Matches → Messages

One-to-Many: each match contains multiple messages.

Users → Messages

One-to-Many (twice):

- as sender: user sends many messages
- as receiver: user receives many messages

5.2 Indexes

- users.email – unique index for fast login lookups
- users.username – unique index
- messages.match_id – index for conversation retrieval
- likes (liker_id, liked_id) – composite unique index

5.3 ERD Diagram

